1. Introduction

A self-balancing binary search tree keeps its height logarithmic in the number of stored keys so that search, insertion, and deletion all run in logarithmic time. The balance is maintained by local restructuring operations called rotations, which preserve the in-order sequence of keys while reducing the height of the affected subtree. A self-balancing binary search tree keeps its height logarithmic in the number of stored keys so that search, insertion, and deletion all run in logarithmic time. The balance is maintained by local restructuring operations called rotations, which preserve the in-order sequence of keys while reducing the height of the affected subtree. A self-balancing binary search tree keeps its height logarithmic in the number of stored keys so that search, insertion, and deletion all run in logarithmic time. The balance is maintained by local restructuring operations called rotations, which preserve the in-order sequence of keys while reducing the height of the affected subtree.

2. Rotations and Invariants

A self-balancing binary search tree keeps its height logarithmic in the number of stored keys so that search, insertion, and deletion all run in logarithmic time. The balance is maintained by local restructuring operations called rotations, which preserve the in-order sequence of keys while reducing the height of the affected subtree. A self-balancing binary search tree keeps its height logarithmic in the number of stored keys so that search, insertion, and deletion all run in logarithmic time. The balance is maintained by local restructuring operations called rotations, which preserve the in-order sequence of keys while reducing the height of the affected subtree. A self-balancing binary search tree keeps its height logarithmic in the number of stored keys so that search, insertion, and deletion all run in logarithmic time. The balance is maintained by local restructuring operations called rotations, which preserve the in-order sequence of keys while reducing the height of the affected subtree.